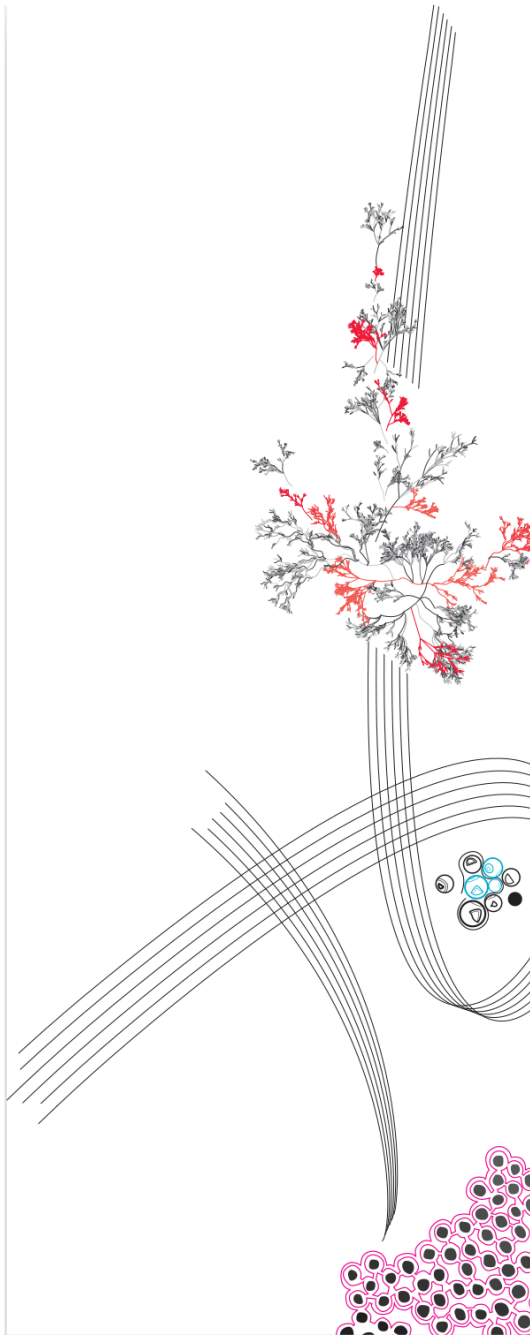




Creative Technology

Module 7 “Hands-on AI”

Project report

Mouth-tracking and targeting

A modular system

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Introduction

Our installation uses a webcam as an input to a U-net model to detect the location of an open mouth and then positions a laser to aim into the open mouth. The system is designed with expandability in mind.

In the current form, the installation does not have much application, however it can be easily adapted to different needs by attaching something instead of the laser. For example, it can be made into a candy dispenser, that would shoot a candy into the mouth.

The model itself can also be applied in different scenarios e.g. helping someone with limited motor skills eat more independently.

The main challenge is to achieve a real-time, accurate detection of an open mouth using a U-net model and using that data as an input to precisely control the laser using two servos and an Arduino.

Hardware

As an input device Creative BlasterX senz3D camera is used. We wanted to use an external webcam with HD capabilities to potentially attach it to the laser station. This camera also has a depth sensor, which we thought could be used for our initial idea of M&M shooter, but since the idea was changed during the project due to time constraints, this feature was not used.

Laptop with a dedicated high end graphics card was used to train and run the model on. This allowed us to train the model in a reasonable time. Additionally, it makes real time prediction faster than a microprocessor would, which is important for us since we work with real time movement.

Arduino Leonardo was used as a microcontroller to drive the servos. This specific brand of the microcontroller was used because it was easily available. The reason why we chose a microcontroller over a microprocessor was because the computations were done on the laptop.

For the actuators we chose the 180° MG995 servo motors for their precise control and because the other available 9g servos are not strong enough for the flexible system we have designed.

The system circuit diagram can be seen in figure 1.

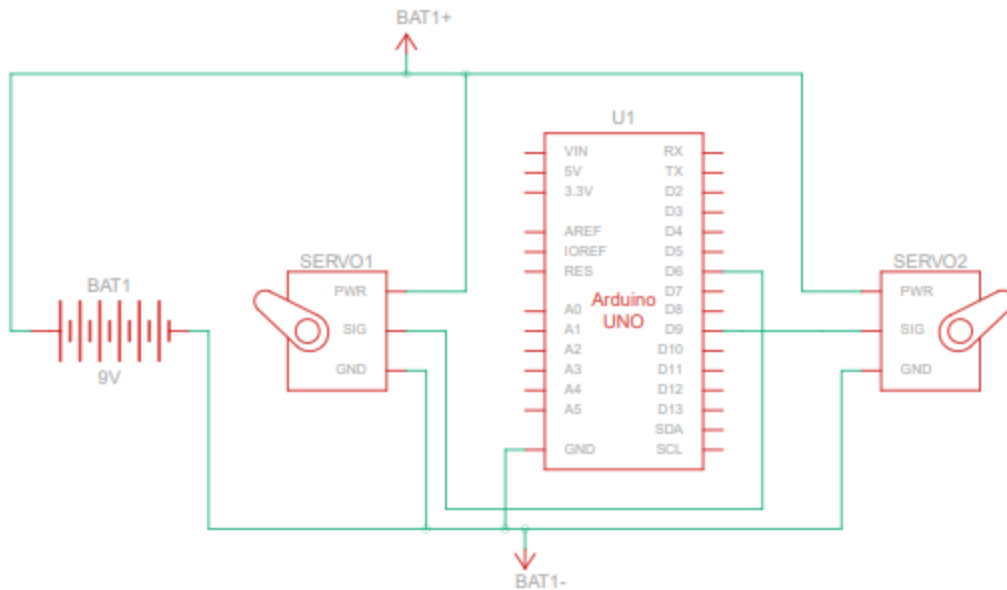


Figure 1 System circuit diagram. Made using TinkerCad

Data collection

Process of data collection

The data collection was approved by the CIS ethics committee of the University of Twente.

To recruit participants, we asked our fellow students to help us with data collection. Before collecting the data, we asked participants to sign the consent form provided by the teacher. Since our model had to detect open mouths, the data we collected were pictures of people with their mouths open and closed. Multiple photos were taken of each participant, from different angles and at different distances from the camera. Various angles and distances in the training dataset improve the model's performance later, on the real-life data. The reason for including pictures of closed mouths is to include negative samples so that the model does not try to find a mouth in every picture. At first, the ratio of open to close mouths was 84 to 14 images respectively. However, after training a model using the initial data, it was often detecting something even when a mouth was closed, especially for people with moustaches. Because of that, we took additional photos of participants who have moustaches with closed mouths.

Preprocessing of data

The dataset was preprocessed by turning the images to grayscale to help the model focus on the shape of the mouth and reducing the distractions that colour would bring in this situation. The faces are then extracted from the images using the OpenCV's implementation of Haar-cascade classifiers. The Haar-cascade is a machine learning-based object detection algorithm that recognises specific structures in an image which are found by comparing the calculated intensity of adjacent regions, this can for example see the difference between the dark spots around eyes and bright foreheads (OpenCV: Face Detection Using Haar Cascades, n.d.). The images were then augmented by introducing gaussian noise to the original image and by brightening and increasing contrast in the original image. Finally, the original, noisy and brightened images are each vertically flipped, rotated by 90° clockwise and 90° counterclockwise. This results in 11 augmentation which means that the original dataset is increased by a factor of 12.

Description of the data set

The final dataset included 122 original images from 24 different participants. The participants were 18–26-year-old CreaTe students. Their gender was not specified but instead we tried to find participants based on the following features: hair length, moustache and beard, glasses, head size and shape.

AI

ML algorithms applied

The task we had was to find the location of an open mouth on the picture and point the laser at it. CNN would only be able to classify if the mouth is open or closed, which was not enough for our goal. Based on this, we decided that we need an image segmentation model, which would be able to find the exact position of the mouth. After some research, we found three potential image segmentation models: U-Net, Fully Connected Network (FCN) and Mask Region-based Convolutional Neural Network (Mask R-CNN). Out of these we chose U-Net because it was initially designed to be used for medical purposes, which makes it better than FCN at detecting fine edges and it works better with smaller datasets. Additionally, it is less complex than Mask R-CNN, which is capable of instance segmentation. But since we did not need this functionality, we opted for a simpler of the two.

A U-Net (Ronneberger et al. 2015) uses convolutional layers in combination with maxpooling and upsampling to extract the features from the inputs (Figure 2). This works the best with image inputs, similar to CNNs. This fits perfectly to our input data type, since we use camera output as the input.

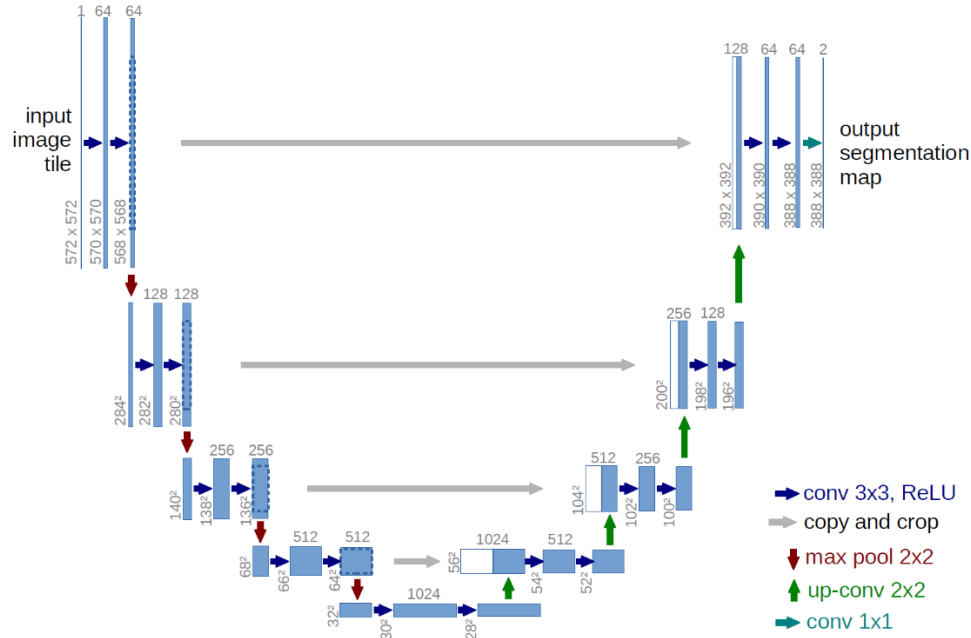


Figure 2 Structure of a general U-Net model (Ronneberger et al. 2015)

The U-Net outputs an array of zeros and ones (mask) where the ones are the detected object. This works great for our need since we can find the middle of the mask and use these coordinates to direct the laser.

Quality of the model

The model was evaluated using the dice coefficient. The dice coefficient is a value between zero and one which describes how well the predicted mask fits the original mask. During the project the dataset was iteratively adjusted which resulted in two working models, the first model had a dice coefficient around 0.8 but based on testing this model had a harder time with detecting closed mouths and open mouths with moustaches. The second model was trained on the updated dataset which resulted in a dice coefficient of 0.7 but it works better on the aspects the first model performed worse on.

Installation

Description of the Installation

Originally, we designed this system as a candy shooter that would shoot e.g. an M&M into your mouth once you open it. Because of the time constraints and miscalculations of the strength of the servos, we changed it to a modular pan and tilt station where a variety of attachments could be mounted on. For this project a laser was mounted on the station to prove that the system works precisely as can be seen in figure 3.

The camera is used as an input device. Using pre trained model we detect if there is a face in a frame and in case there is, crop it and input to the U-Net model. The model detects where the open mouth is and outputs its mask. Using moments of the output mask, we find a centre of the mouth and combine it with the position of the cropped part of the face to find the coordinates of the centre in the original frame, relative to the horizontal and vertical middle lines. This position is sent to the Arduino microcontroller where it gets mapped to an angular position which is sent to two servo motors. Finally, the servos point the attached laser at the open mouth.



Figure 3 Laser station

Evaluation

The installation was tested with us and a person that was not present in the dataset. During the first testing, we noticed that the model often detects open mouth when it is not present and that it performs poorly on people with moustaches. Because of that, additional photos of closed mouths and faces with moustaches were taken to better train the model.

The laser station reacts well to the output of the model. However, the way it is designed right now, to have the laser point directly at the mouth, it needs to be placed at the same height and distance from the person as the camera. This could be prevented by attaching an external camera to the base of the laser station.

Conclusion

At the end of this project, a working system was created. The initial idea was to make a M&M shooter but not being able to design a working shooting mechanism prevented us from achieving it. Working on a new type of model (image segmentation model) and making that interact with the actuators was interesting. Throughout this process we learned more about data collection and data preprocessing for deep learning models, for example the importance of negative samples (in this case images with closed mouths).

There are still couple of things we wanted to implement, but due to the time limit, we could not have done that. One of them is making the model simpler. During the training we noticed that some of the convolutional layers do not output any features, which could mean that they can be removed. This would improve the efficiency of the model. The other thing we would have liked to do is increasing the speed of the predictions, thus increasing the framerate. This could be done by running the model on the GPU. In our case this was not possible to implement this due to camera driver problems on WSL, this could be resolved by running the model in a native linux or macOS environment.

Bibliography

Ronneberger, O., Fischer, P., & Brox, T. (2015). U-NET: Convolutional Networks for Biomedical Image Segmentation. In *Lecture notes in computer science* (pp. 234–241). https://doi.org/10.1007/978-3-319-24574-4_28

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